

## Object-Oriented Software Engineering - CA Part 2 Report

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Business Domain: Beauty Salon System

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### Task 8 – Agile Methodology Proposal and Comparison

For the development of the Beauty Salon System, we adopted the Scrum Agile framework. This approach supports iterative development, adaptability to change, and ongoing customer feedback, all of which are critical in a dynamic domain such as personal services.

#### Why Scrum for a Beauty Salon System?

**Flexibility:** Changes in services, pricing, or customer preferences can be quickly adapted.

**Iterative Delivery:** Each sprint focused on a core feature: booking, staff views, services, etc.

**Client Involvement:** Regular feedback sessions allow early corrections.

**Transparency:** Daily stand-ups, backlogs, and sprint reviews improve visibility.

#### Scrum vs Waterfall:

| Feature         | Scrum (Agile)                | Waterfall            |
|-----------------|------------------------------|----------------------|
| Project Flow    | Iterative and incremental    | Linear, sequential   |
| Flexibility     | High, change-tolerant        | Low, change-averse   |
| Risk Management | Early detection via feedback | Late-stage detection |
| Delivery Style  | Frequent working releases    | One final release    |

Scrum allows us to build features in manageable chunks, resulting in faster feedback cycles and fewer late-stage surprises.

### Task 9 – Agile Artifacts

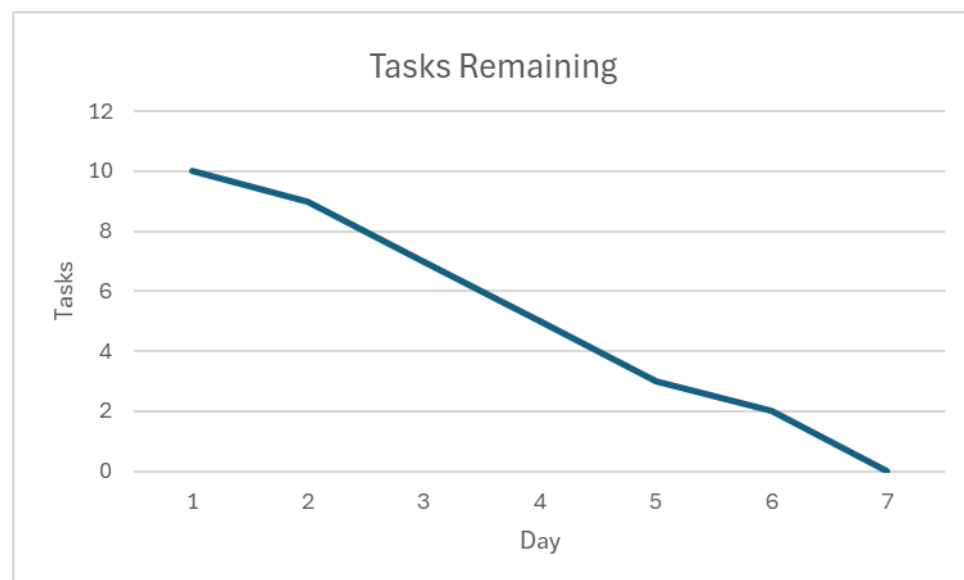
## User Stories:

| User Story ID | As a...      | I want to...                       | So that I can...                        | Priority |
|---------------|--------------|------------------------------------|-----------------------------------------|----------|
| US01          | Customer     | Book an appointment online         | Choose a convenient time slot           | High     |
| US02          | Admin/Staff  | View and manage daily appointments | Know which stylist is assigned and when | High     |
| US03          | Staff Member | Mark an appointment as complete    | Track finished vs pending jobs          | Medium   |
| US04          | Admin        | Add or remove services             | Keep the service menu updated           | Medium   |
| US05          | Customer     | Receive an SMS/email reminder      | Not forget my appointment               | Low      |

## Product Backlog:

| Sprint | Backlog Items                        | Status      |
|--------|--------------------------------------|-------------|
|        | 1 User login + role setup            | Completed   |
|        | 1 Appointment booking (basic)        | Completed   |
|        | 2 Appointment view/edit (Admin view) | In progress |
|        | 2 Service list management            | In progress |
|        | 3 Email reminders                    | Not Started |
|        | 3 Payment integration (placeholder)  | Not Started |

## Burndown Chart:



## Task 10 – Risk, Quality & Communication Management

### Project Risks

| Risk                             | Likelihood | Impact | Mitigation Strategy                            |
|----------------------------------|------------|--------|------------------------------------------------|
| Feature creep from stakeholders  | Medium     | High   | Lock requirements before each sprint           |
| Inconsistent testing across team | Medium     | Medium | Implement TDD and write test cases in parallel |
| Missed deadlines                 | Low        | High   | Use Trello for sprint tracking & time-boxing   |
| Version conflicts                | Medium     | Medium | Use GitHub with clear commit messages          |

## Quality Strategy

TDD: We wrote tests before implementation.

Code Reviews: Peer reviews for each class.

UML Alignment: All implementation matched UML diagrams.

Validation: Used assertions to verify expected outcomes.

## Communication Strategy

Google Meet: Weekly Scrum meetings

Google Docs: Shared documentation repo

## Task 11 – Class Implementation Overview

We implemented the Appointment Booking Use Case. The Java classes created include:

`Appointment.java` – data model representing bookings

`BookingManager.java` – controller class to add appointments and check for time conflicts

## Structure

`src/model/Appointment.java`

`src/controller/BookingManager.java`

## Task 12 – Testing Strategy and Results

We adopted Test-Driven Development (TDD using JUnit 5).

## Test Files

`TestAppointment.java` – Verifies object construction and getters

`TestBookingManager.java` – Tests adding appointments and checking for time conflicts

#### JUnit Result

```
*
+-- JUnit Jupiter [OK]
| +-- TestAppointment [OK]
| | '-- testAppointmentCreation() [OK]
| '-- TestBookingManager [OK]
|   +-- testAddAppointment() [OK]
|   '-- testConflict() [OK]
+-- JUnit Vintage [OK]
'-- JUnit Platform Suite [OK]

Test run finished after 106 ms
[      5 containers found      ]
[      0 containers skipped    ]
[      5 containers started    ]
[      0 containers aborted    ]
[      5 containers successful ]
[      0 containers failed     ]
[      3 tests found           ]
[      0 tests skipped         ]
[      3 tests started         ]
[      0 tests aborted         ]
[      3 tests successful      ]
[      0 tests failed          ]
```

All tests executed successfully via the standalone JUnit Console.

#### Declaration

I hereby declare that this assignment is entirely my own work and has not been plagiarised in part or in whole.

Signed: James Canning

Date: 29/05/25